

CAB330: Assignment 2 Specification

(Descriptive Data Mining)

Introduction

The purpose of this assignment is to develop your understanding of how the methods learned in this unit can be applied to different types of datasets, including structured record data, transactional data, unstructured text data, and web log data. The assignment includes four tasks for the group component, covering Clustering, Association Mining, Text Processing and Analysis, and Web Mining. You can use Python and the libraries and techniques introduced throughout the unit.

This assignment comprises both a **group component** (Tasks 1–4, described in this document) and an **individual component** (see the Canvas site for further details). The specific tasks for the group component are outlined below.

Group Component (Tasks 1 to 4)

Task 1: Unsupervised Learning - Clustering (5 Marks)

The dataset **Dataset4Task1_26.csv** contains physicochemical properties of white wine samples from the north of Portugal. It includes various physicochemical attributes of white wine, with each row representing a single white wine sample. Each sample is described by the following eleven attributes (variables):

Attribute	Description
fixed acidity	This refers to the acids that are not volatile, such as tartaric acid. It plays a significant role in the taste and stability of the wine. Higher fixed acidity often contributes to a sharper taste.
volatile acidity	This is the amount of acetic acid present in wine. Higher levels of volatile acidity can lead to a vinegar-like taste and are generally considered a fault in wine.
citric acid	Adds freshness and flavor to the wine. It's a minor acid in wine but can contribute to the overall taste and stability.
residual sugar	The amount of sugar remaining after fermentation. It can influence the sweetness of the wine. Wines with higher residual sugar are sweeter, while those with less are drier.
chlorides	Refers to the amount of salt in the wine, which can affect the taste. High levels can lead to a salty taste.
free sulfur dioxide	This acts as an antioxidant and preservative, helping to protect wine from spoilage and oxidation. It is the portion of sulfur dioxide that is not bound to other compounds and is available to act as a preservative.
total sulfur dioxide	The sum of free and bound sulfur dioxide. It plays a crucial role in preventing oxidation and microbial spoilage.
density	Related to the sugar content of the wine. Higher density indicates higher sugar content, which can be linked to residual sugar and alcohol content.
pH	Measures the acidity or alkalinity of the wine. Lower pH values indicate higher acidity, which can contribute to the wine's crispness and stability.
sulphates	Add to the wine's SO ₂ content and act as a preservative. They can also influence the mouthfeel and overall quality of the wine.

alcohol	The ethanol content of the wine, usually expressed as a percentage. It affects the body, flavor, and warmth of the wine.
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Assume that a winery wants to categorise its white wine samples using k -means clustering. Your task is to apply k -means clustering to identify the **minimum effective number of clusters** and then describe and interpret the characteristics of each resulting cluster.

Answer the following questions based on the dataset and your clustering analysis.

- 1) Report the number of missing values in each column of the dataset (Dataset4Task1_26.csv). Describe and justify the data-cleaning approach adopted, including the appropriate preprocessing steps used to produce a clean dataset suitable for subsequent analysis.
- 2) Use all variables in the dataset to construct a default K -means clustering model, aiming to partition the data into more than two clusters. Determine the optimal number of clusters by applying both the Elbow Method, by plotting K (2 to 16) against the within-cluster sum of squares (inertia); and the Silhouette Scores. Compare and interpret the results from both approaches to justify the selection of the most appropriate K .
- 3) For high-dimensional datasets, clustering results can be difficult to interpret when many features are used. Describe your approach for selecting a smaller set of relevant features to improve clustering interpretability and identify the features you selected.
- 4) Based on your selected variables, create a clustering model with the optimal K value you discovered in 2). Afterward, address the following sub-questions:
 - a) Identify the number of records assigned to each cluster, report the centroid of each cluster, and calculate the Silhouette Score.
 - b) Plot the cluster distribution using a *pairplot* and explain the key characteristics based on the feature distributions within each cluster, the degree of cluster separation (i.e., which pairs of features best distinguish the clusters), and the extent of cluster overlap.
- 5) Use `StandardScaler` and `MinMaxScaler` from the `sklearn` package to transform the feature values, then develop two additional models based on the model developed in 4).
 - a) Compare the two models and identify which transformation method performs better.
 - b) Use Seaborn functions (e.g., `histplot` and `kdeplot`) to interpret the clustering results of the better-performing model identified in a), focusing on feature distributions and cluster separation.

Task 2: Unsupervised Learning - Association Mining (5 Marks)

A bakery is interested in identifying associations among items purchased by its customers. To achieve this, the bakery has chosen to conduct an association mining using the transaction dataset **Dataset4Task2_26.csv**. Details of the dataset are provided below:

Attribute	Description
TransactionNo	Unique identifier for each individual transaction.

Items	The name of the purchased product.
DateTime	Date and time stamp of the transactions.
Daypart	Part of the day when a transaction is completed, including morning, afternoon, evening, and night.
DayType	It determines whether a transaction occurred on a weekend or a weekday.

Your task is to conduct an association analysis on this dataset and answer the following questions based on the dataset and your association mining model.

- 1) Observe and clean the data: Inspect the data in the **Dataset4Task2_26.csv** file by reporting the number of unique products and checking for missing (NaN) values. Then, identify and address data quality issues that may affect the transformation of the data into a transaction-item representation. Consider issues such as missing values, invalid transaction IDs, or other relevant data quality problems. Describe your data-cleaning approach and apply the appropriate preprocessing steps to produce a clean dataset suitable for association rule mining.
- 2) Apply the Apriori algorithm to conduct association rule mining and answer the following questions:
 - a) Select the appropriate variables to transform the clean dataset into a transaction-item representation. Use $min_sup = 0.02$ and $min_conf = 0.1$ and apply the Apriori algorithm to identify the frequent itemsets and generate association rules in the following format:
“Left_side Right_side Support Confidence Lift”
 - b) What is the highest confidence value among the resulting association rules? Which rule has this value? And what is the highest lift value among the resulting association rules? Which rule has this value?
- 3) Plot the support, confidence, and lift values for all resulting association rules. Then, identify and plot the association rules with $lift \geq 1$, and clearly display the corresponding Left_side and Right_side item names. At last, interpret three positive association rules from the resulting rule set.
- 4) The store is particularly interested in products that individuals purchase when they buy “Pastry”.
 - a) How many rules are in the subset?
 - b) Based on the rules, what are the other products these individuals are most likely to purchase?
- 5) If the store wants to boost the sales of "Medialuna", which products should be placed nearby based on the discovered association rules? Justify your answer using your knowledge of association mining and explain how the relevant rule(s) can support this decision-making process.

Task 3: Text processing and analysis (5 Marks)

An online bookshop is planning to provide an online recommendation service to its customers. The shop has collected a metadata dataset, **Dataset4Task3_26.csv**, which contains information about a collection of books. In this task, you are required to perform text processing and analysis on the dataset to identify clusters of books based on similar topics and themes derived from their descriptions. The details of the dataset are provided below.

Attribute	Description
Title	Book title
description	The description of book
authors	Name of book authors

publisher	Name of the publisher
publishedDate	The date of publish
categories	Genres of books

Answer the following questions based on the dataset and your analysis.

- 1) Design a text-preprocessing function (or lemmatisation process) to preprocess the text in the **description** column. The function should split each description into lowercase tokens and remove unwanted tokens. Clearly define the types of unwanted tokens to be removed, such as stop words, punctuation, numbers, and other non-informative tokens. Apply the function to the dataset and print the first 10 processed descriptions to verify that the preprocessing function works correctly. Then, calculate and report the average text length of the **description** column, based on the number of tokens in each processed description.
- 2) Design a TF-IDF calculation function that returns a list of terms along with their term frequency (TF) and document frequency (DF) for a given text. Then, design a function to generate a similar Zipf plot for the processed **description** data to show the term frequency ranking vs its DF.
 - a) Test the two functions using the processed **description** data and verify that they produce the expected outputs.
 - b) Based on the Zipf plot, identify and list the 12 most commonly occurring terms across the description. Explain why these terms may be less useful for text analysis, particularly for document clustering.
- 3) Develop K-Means Clustering Model 1 using the vectorised text features without applying feature selection or document-frequency (DF) filtering.
 - a) Vectorisation: Use `TfidfVectorizer()` to transform the *description* field into TF-IDF-weighted feature vectors.
 - b) K-Means clustering Model 1: Develop a K-Means clustering model with 6 clusters. Examine the top 10 terms associated with each cluster and assign a *meaningful name* to each cluster based on these terms.
- 4) Develop K-Means Clustering Model 2 using the vectorized text features with feature selection or document-frequency (DF) filtering to removes very rare and very common terms
 - a) Vectorisation: Use `TfidfVectorizer(min_df=3, max_df=0.7)` to transform the *description* field into feature vectors.
 - b) K-Means clustering Model 2: Develop a K-Means clustering model with 6 clusters. Examine the top 10 terms associated with each cluster and assign a *meaningful name* to each cluster based on these terms.
- 5) Use at least two evaluation methods (e.g., top-term analysis and comparison with the 'category' field, SVD or PCA visualization, or Silhouette score) to compare the clustering models and recommend the better-performing model. Then, explain how the resulting clusters could be used in the planned online personalized book recommendation service.

Task 4: Web Log Analysis (5 Marks)

An e-commerce company has collected a web server log file, **Dataset4Task4_26.txt**, which contains information about events recorded on its website. Please note that a '-' in a field

indicates missing data. The log file provides information about website visitors, their browsing behaviour, and crawlers accessing the website, among other activities. In Task 4, you are required to analyse the log file to gain insights into website users' browsing patterns and use the discovered patterns to provide data-driven decision making for the company.

The following table shows the attributes (columns) recorded in the log file.

Attribute	Data Type	Description
ip	String	Client's IP address
auth	String	Remote name of the User performing the request.
ident	String	The ID of the user making the request.
time	Date	The date and time of the request are represented in UTC format as follows: Day/Month/Year: Hour:Minutes:Seconds +Time-Zone-Correction.
method	Categorical	The type of request (GET, POST, PUT, DELETE) that the server received.
path	String	The path of the website related to the request.
proto	String	Protocol used for connecting to the server and its version.
status	Integer	The server returns a status code for each request.
Size	Integer	The data in bytes that was sent back to the client.
referrer	String	The source of the user's referral to the current website, represented as "-" if there's none.
user_agent	String	The user agent string provides details of the browser and host device, such as the name, version, and device type.

Your task is to pre-process the given dataset and apply a suitable unsupervised learning technique, such as association mining or clustering, to the raw web log data. Answer the following questions based on the dataset and the unsupervised learning technique you have selected.

- 1) Pre-process the web log data to identify and extract useful attributes from the columns in the text file. The pre-processing should include:
 - a) Reading the web log file line by line.
 - b) Parsing each log entry to extract the relevant attributes from the raw log data.
 - c) Converting the parsed log entries into a Pandas DataFrame.
 - d) Displaying the resulting DataFrame in a table format similar to the example provided below:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14999 entries, 0 to 14998
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   ip              14999 non-null  object
1   ident          14999 non-null  object
2   auth           14999 non-null  object
3   time           14999 non-null  object
4   method         14999 non-null  object
5   path           14999 non-null  object
6   proto          14999 non-null  object
7   status         14999 non-null  object
8   size           14999 non-null  object
9   referrer       14999 non-null  object
10  user_agent     14999 non-null  object
dtypes: object(11)
memory usage: 1.3+ MB
```

- 2) Implement the following data-cleaning steps to prepare the dataset for analysis, focusing specifically on web page requests:

- a) Convert the text-based `time` column into a proper Pandas `datetime` object.
 - b) Remove the original attributes that contain a large proportion of missing values or are no longer required for the analysis.
 - c) Sort the remaining records in descending order based on the `datetime` column and then display the first 10 rows of the sorted DataFrame.
- 3) Implement the following steps to focus the analysis on webpage/document requests rather than static media resources:
 - a) Remove all requests for image and audio files from a web-server log dataset.
 - b) Determine and report the number of rows (objects) containing requests for images, and the number of rows (objects) containing requests for audio.
- 4) Define and identify user sessions in the web log data. A *user session* is a sequence of web requests made by the same user during a defined period of activity on a website, such as a 30-minute session timeout.
 - a) Provide a clear definition of a user session to be used for Task 4, including the criteria used to determine when a new session begins.
 - b) Develop code to identify user sessions in the web log data and add the following columns to the dataset:
SessionID (a unique identifier assigned to each user session); and **Steps** (the sequential number of each web request within a session, representing the user's navigation steps during that session).

Use the sample output below as a reference for the expected result:

	ip	method	path	proto	status	Datetime	SessionID	Steps
0	54.36.149.41	GET	/filter/27/13%20%D9%85%DA%AF%D8%A7%D9%BE%DB%8C...	HTTP/1.1	200	2019-01-22 03:56:14	1	1
1	31.56.96.51	GET	/image/60844/productModel/200x200	HTTP/1.1	200	2019-01-22 03:56:16	2	1
2	31.56.96.51	GET	/image/61474/productModel/200x200	HTTP/1.1	200	2019-01-22 03:56:16	2	2
3	40.77.167.129	GET	/image/14925/productModel/100x100	HTTP/1.1	200	2019-01-22 03:56:17	3	1
4	91.99.72.15	GET	/product/31893/62100/%D8%B3%D8%B4%D9%88%D8%A7%...	HTTP/1.1	200	2019-01-22 03:56:17	4	1
5	40.77.167.129	GET	/image/23488/productModel/150x150	HTTP/1.1	200	2019-01-22 03:56:17	3	2
6	40.77.167.129	GET	/image/45437/productModel/150x150	HTTP/1.1	200	2019-01-22 03:56:18	3	3
7	40.77.167.129	GET	/image/576/article/100x100	HTTP/1.1	200	2019-01-22 03:56:18	3	4
8	66.249.66.194	GET	/filter/b41,b665,c150%7C%D8%A8%D8%AE%D8%A7%D8%...	HTTP/1.1	200	2019-01-22 03:56:18	5	1
9	40.77.167.129	GET	/image/57710/productModel/100x100	HTTP/1.1	200	2019-01-22 03:56:18	3	5
10	207.46.13.136	GET	/product/10214	HTTP/1.1	200	2019-01-22 03:56:18	6	1
14	91.99.72.15	GET	/product/10075/13903/%D9%85%D8%A7%DB%8C%DA%A9%...	HTTP/1.1	200	2019-01-22 03:56:19	4	2
11	40.77.167.129	GET	/image/578/article/100x100	HTTP/1.1	200	2019-01-22 03:56:19	3	6
12	178.253.33.51	GET	/m/product/32574/62991/%D9%85%D8%A7%D8%B4%DB%8...	HTTP/1.1	200	2019-01-22 03:56:19	7	1

- 5) Apply an appropriate unsupervised learning method (e.g., association mining or clustering) to the processed dataset developed in 4).
 - a) Select an appropriate unsupervised learning method and justify the rationale for your choice.
 - b) Identify and explain the variables/features included in the analysis and justify their selection.
 - c) Present a case study demonstrating how the discovered patterns or clusters can be used to support data-driven decision making.